



Cloud Data Centers: Finding the Balance Between Energy Savings and System Speed

CSCI 6093 – Final Project

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Introduction

Artificial Intelligence (AI) cloud computing infrastructures are known to consume massive amounts of energy to operate.

As national infrastructure continues to build at a rapid pace, the power required to maintain these servers is hitting unprecedented levels.

Reducing this energy footprint is important not only for environmental sustainability, but also for the economic viability of the businesses that rely on these services.

Objectives



Evaluating the trade-offs in power, efficiency and cost in cloud infrastructures.

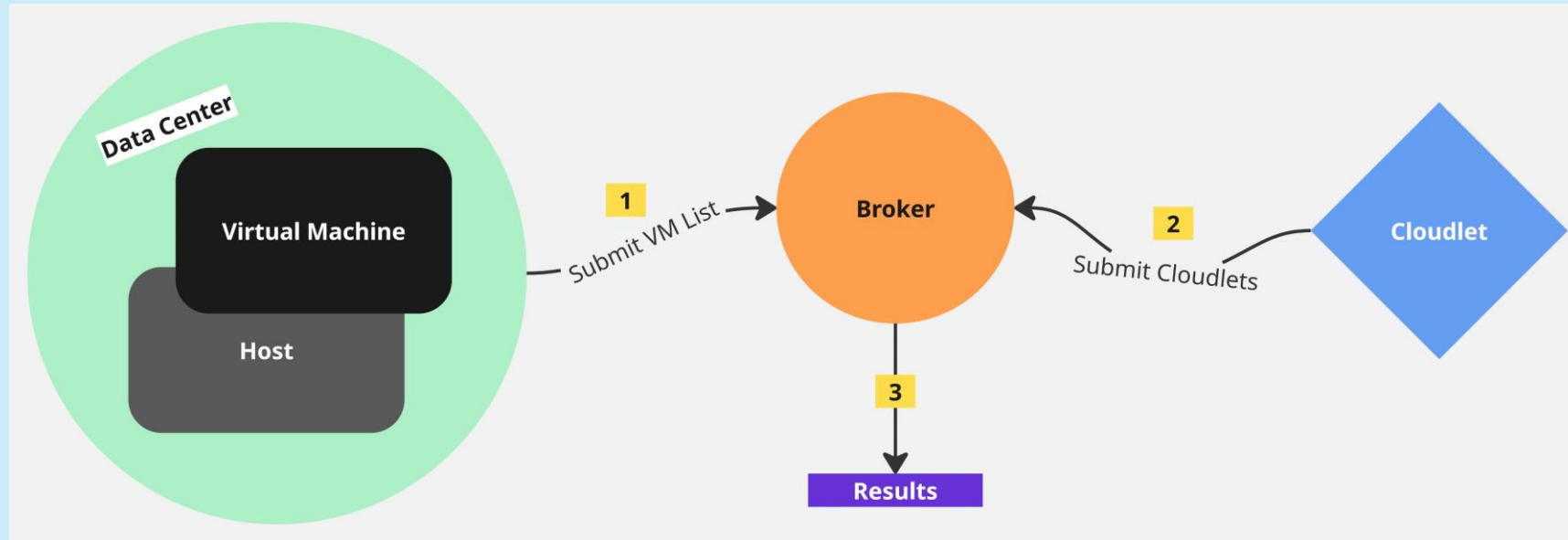


Learning about CloudSim Plus, a framework for creating simulations of cloud computing infrastructures.

Background

- The total power needed to keep data centers running is reaching record highs, in a “race to build national AI infrastructure.”
- By 2030, AI data centers will account for 3% of total global electricity consumption.
- **CloudSim Plus**
 - A widely used simulation framework in cloud computing research
 - Popular fork of CloudSim, the first open-source specialized cloud simulation framework
 - Users can specify parameters about the hardware

CloudSim Plus



Methodology

- A comparison of 2000s, 2010s, and 2020s data centers, using CloudSim Plus to simulate each type
- Collected specs for each server type, from SPECpower_ssj2008 benchmarks
- Each data center has the same number of hosts (servers), and the same number of tasks to complete
 - 100 hosts each
 - 10,000 tasks each with a length of 50,000 MI (Millions of Instructions)
- 3 million rows of results then exported as a CSV file and analyzed in Power BI

Methodology - Legacy Data Center

- **HP ProLiant DL360 G1**
 - Enterprise Server
 - Benchmarked in 2000
- Specs:
 - **RAM:** 2,048 MB
 - **Storage:** 36,000 MB
 - **BW:** 10 Mbps
 - **PE / Cores (x2):** 1,000 MIPS each
 - **Active power draw:** 292 W
 - **Idle power draw:** 204 W



Methodology - Modern Data Center

- **HPE ProLiant DL360 Gen10**
 - Enterprise Server
 - Benchmarked in 2017
- Specs:
 - **RAM:** 98,304 MB
 - **Storage:** 340,000 MB
 - **BW:** 4,000 Mbps
 - **PE (x56):** 2,500 MIPS each
(two processors with 28 cores each)
 - **Active power draw:** 451 W
 - **Idle power draw:** 48 W



Methodology - AI Data Center

- **HPE ProLiant DL380a Gen11**
 - Enterprise Server
 - Benchmarked in 2024
- Specs:
 - **RAM:** 262,144 MB
 - **Storage:** 122,880,000 MB
 - **BW:** 4,000 Mbps
 - **PE (x128):** 15,000 MIPS each
(two processors with 64 cores each)
 - **Active power draw:** 736 W
 - **Idle power draw:** 235 W



Energy Evaluation Metrics

OPERATIONAL COST

- **Total energy consumption (E) =**
(Execution time × Max Active Power) × 100
 - Legacy = 1,460,000 J
 - Modern = 902,000 J
 - AI = 220,800 J
- **Cost = $E \times \text{Rate}$**
 - GA Power Commercial Rate: estimated at **\$0.115 per kilowatt-hour**

HARDWARE COST

- **Power Usage Effectiveness =**
Data center energy / Server energy
 - Legacy = 2.5 PUE
 - Modern = 1.58 PUE
 - AI = 1.56 PUE
- Estimated physical hardware costs
 - Legacy = \$56,000
 - Modern = \$650,000
 - AI = \$15,000,000



Conclusion

- The AI data center demonstrated superior energy efficiency, but was offset by significantly higher hardware cost
- The modern data center is the most practical middle ground, at least while AI hardware costs remain exorbitant
- Raw energy consumption alone is a misleading metric
- Future work can:
 - Incorporate software and environmental costs
 - Apply server cost amortization for a more long-term financial picture
 - Use the *schedulers* class to add server idle times
 - Use the *addOnCloudletFinishListener()* method to allow dynamic task submission

